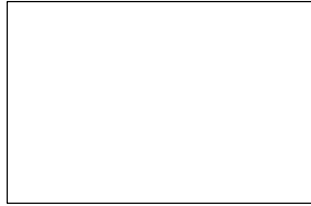


Power



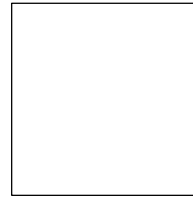
File: power.kicad_sch

Drivers



File: drivers.kicad_sch

Mosfets



File: mosfets.kicad_sch

BRAKE



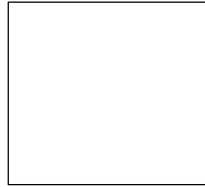
File: brake.kicad_sch

Shunt_Current_Sense



File: shunt_current_sense.kicad_sch

VBus_Current_Sense



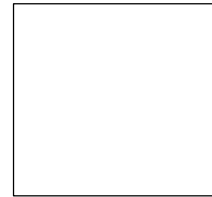
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MCU



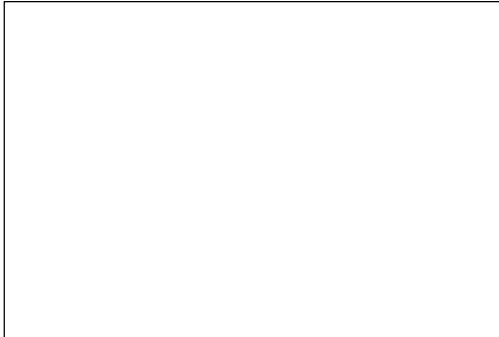
File: mcu.kicad_sch

HALL_ENC



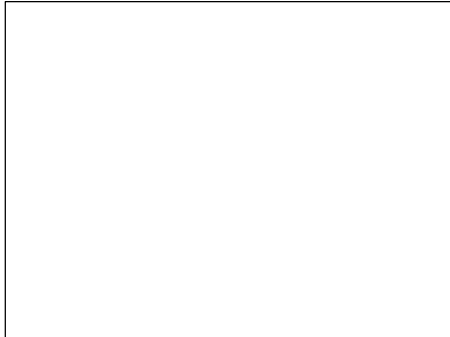
File: hall_enc.kicad_sch

Non_Isolated_Sources



File: Non_Isolated_Sources.kicad_sch

Isolated_Sources

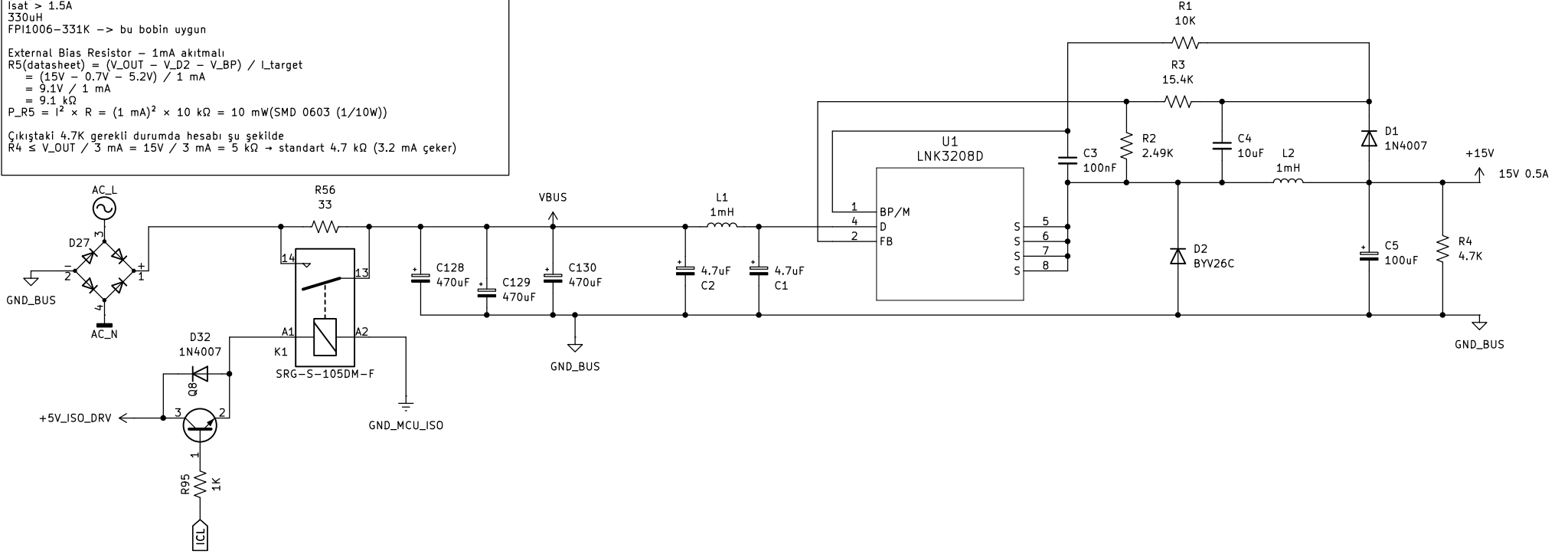


File: Isolated_Sources.kicad_sch

Çıkış bobini:
 $I_{sat} > 1.5A$
 $330\mu H$
 FPI1006-331K -> bu bobin uygun

External Bias Resistor - 1mA akıtmalı
 $R5(datasheet) = (V_{OUT} - V_{D2} - V_{BP}) / I_{target}$
 $= (15V - 0.7V - 5.2V) / 1mA$
 $= 9.1V / 1mA$
 $= 9.1k\Omega$
 $P_{R5} = I^2 \times R = (1mA)^2 \times 10k\Omega = 10mW(SMD 0603 (1/10W))$

Çıkıştaki 4.7K gerekli durumda hesabı şu şekilde
 $R4 \leq V_{OUT} / 3mA = 15V / 3mA = 5k\Omega \rightarrow standart 4.7k\Omega (3.2mA \text{ çeker})$



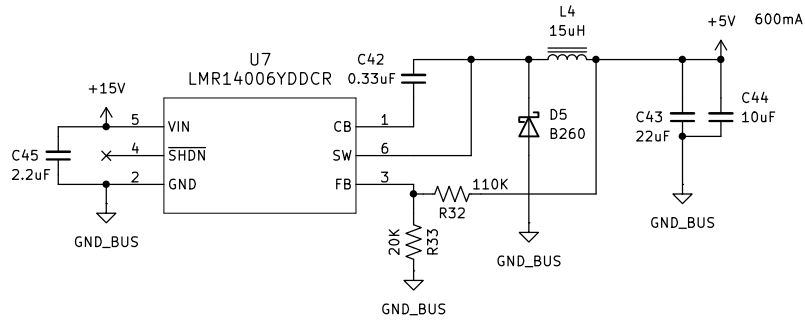
$$C = 2P / (\Delta V \times V_{bus} \times 2\pi f) \rightarrow \%10 \text{ ripple}(31V) 60Hz, 310V$$

$$R = V_{bus} / I_{hedef_max} = 310 / 10A = 31\Omega \rightarrow 33\Omega \text{ standart de\u011fer}$$

$$E = \frac{1}{2} \times C \times V^2 = \frac{1}{2} \times 1000\mu F \times 310^2 = 48J \text{ (\u015farj s\u00fcresince diren\u00e7 \u00fczerinde)}$$

$$\tau = R \times C = 33 \times 1000\mu F = 33ms$$

Burası biraz daha incelenecek



UCC217 Datasheet→ Turn-On and Turn-Off Gate Resistors altında hesaplar mevcut

$R_{OH_EFF} = 0.7 \, \Omega$ (driver pull-up, switching transient)

$R_{OL} = 0.3 \, \Omega$ (driver pull-down)

$R_{G_int} = 6 \, \Omega$ (C3M0120065J internal)

$\Delta V = V_{DD} - V_{EE} = 20V$

Peak Gate Akımları

$$\begin{aligned} I_{source_pk} &= \min(10A, \Delta V / (R_{OH_EFF} + R_{g_on} + R_{G_int})) \\ &= \min(10A, 20 / (0.7 + 22 + 6)) \\ &= \min(10A, 20/28.7) \\ &= 0.70 \, A \end{aligned}$$

$$\begin{aligned} I_{sink_pk} &= \min(10A, \Delta V / (R_{OL} + R_{g_off} + R_{G_int})) \\ &= \min(10A, 20 / (0.3 + 10 + 6)) \\ &= \min(10A, 20/16.3) \\ &= 1.23 \, A \end{aligned}$$

Sürücü İç Güç Tüketimi (Termal Limit Kontrolü)

$$P_Q \text{ (quiescent)} = 5mA \times 20V = 100 \, mW$$

$$P_{SW} \text{ (switching)} = 0.5 \times 0.0428 \times 20V \times 20kHz \times 26nC \approx 0.22 \, mW$$

$$P_{DR} \text{ (toplam)} \approx 100 \, mW$$

$$\begin{aligned} 0.0428 &= R_{OH_EFF} / (R_{OH_EFF} + R_{g_on} + R_{G_int}) + R_{OL} / (R_{OL} + R_{g_off} + R_{G_int}) \\ &= 0.7 / (0.7 + 22 + 6) + 0.3 / (0.3 + 10 + 6) \\ &= 0.7 / 28.7 + 0.3 / 16.3 \\ &= 0.0244 + 0.0184 \\ &= 0.0428 \end{aligned}$$

$$P_{Rg_on} = (22/28.7) \times 0.5 \times 26nC \times 20V \times 20kHz \approx 4.0 \, mW$$

1/8W direnç

$$R_{g_on} = 22 \, \Omega, R_{g_off} = 10 \, \Omega$$

$$P_{Rg_off} = (10/16.3) \times 0.5 \times 26nC \times 20V \times 20kHz \approx 3.2 \, mW$$

AIN Pin kullanımı, MCU tarafında APWM filtreden geçirilir ve ADC'ye verilir.

Daha kolay gibi, bu yöntemi tercih edeceğim, hesap zinciri →

1. dereceden basit

$f_c \ll 400 \, kHz$ (PWM dalgalanması kalmasın) → en fazla ~40 kHz

$f_c \gg$ sinyal BW (sıcaklık yavaş, ~10–100 Hz yeterli)

Sweet spot: $f_c \approx 1\text{--}10 \, kHz$

$$R_{filt} = 10 \, k\Omega$$

$$C_{filt} = 10 \, nF$$

$$f_c = 1 / (2\pi \times 10k \times 10n) \approx 1.6 \, kHz$$

U_High_Mosfet_Driver



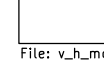
File: u_h_mosfet_driver.kicad_sch

U_Low_Mosfet_Driver



File: u_l_mosfet_driver.kicad_sch

V_High_Mosfet_Driver



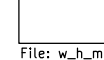
File: v_h_mosfet_driver.kicad_sch

V_Low_Mosfet_Driver



File: v_l_mosfet_driver.kicad_sch

W_High_Mosfet_Driver



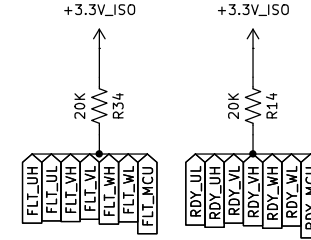
File: w_h_mosfet_driver.kicad_sch

W_Low_Mosfet_Driver



File: w_l_mosfet_driver.kicad_sch

FLT_MCU pini error triger olarak MCU'ya gidecek.
Böylece Donanımsal Driver Protection olacak



DESAT Blanking Süresi Analizi

Formül Hesabı

Devrendeki parametreler:

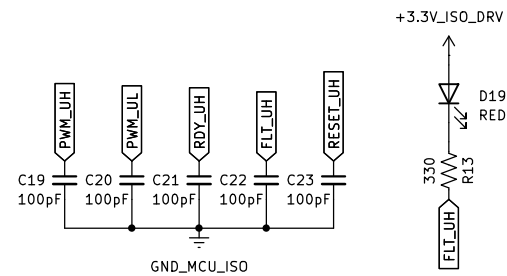
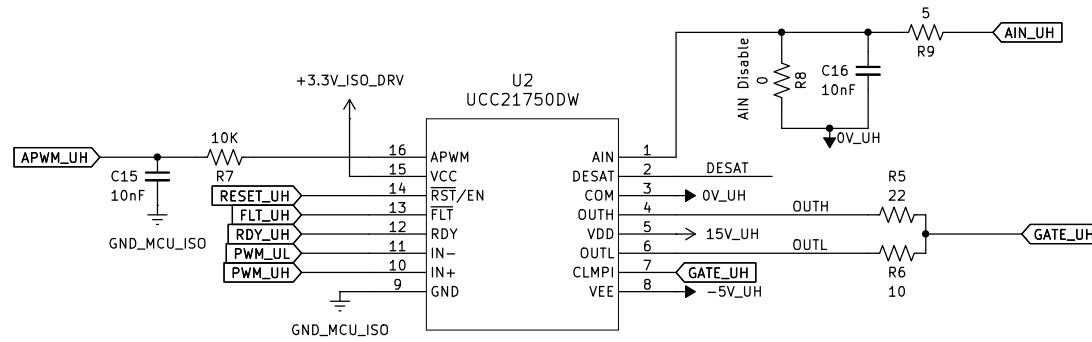
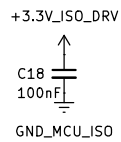
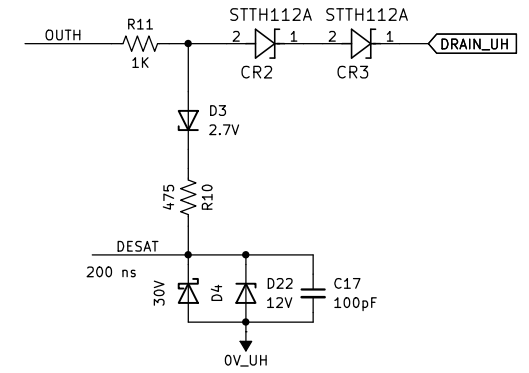
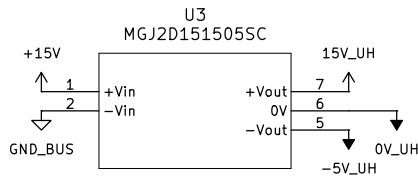
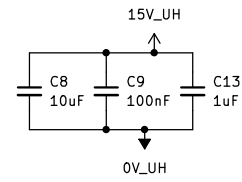
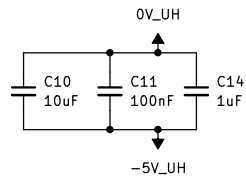
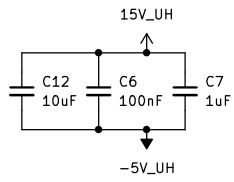
- $V_{DD_eff} = V_{OUTH} - V_{f(D3)} = 15 - 1 = 14 \, V$
- $R_{chg} = R_{11} + R_{10} = 1k + 475 = 1475 \, \Omega$ (Rlim datasheet'te belirtilmemiş, 0 alın)
- $C_{BLK} = C_{17} = 100 \, pF$
- $I_{CHG} = 500 \, \mu A$ (typ), $V_{DESAT} = 9.15 \, V$ (typ)
- $\tau = R_{chg} \times C_{BLK} = 1475 \times 100p = 147.5 \, ns$

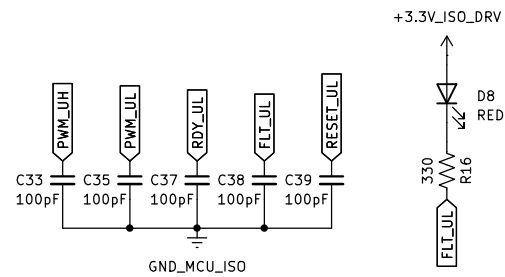
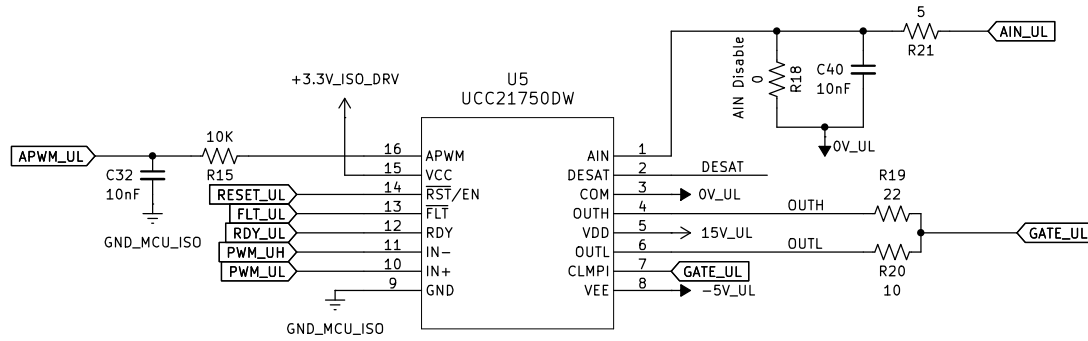
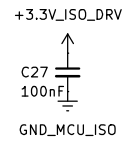
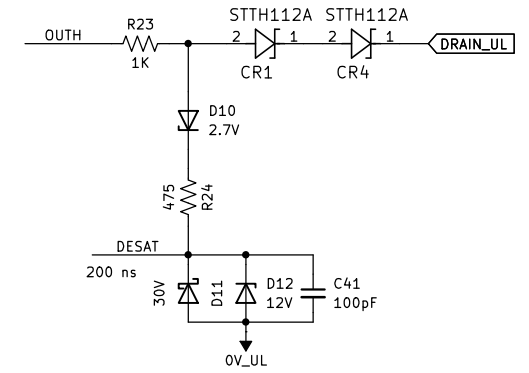
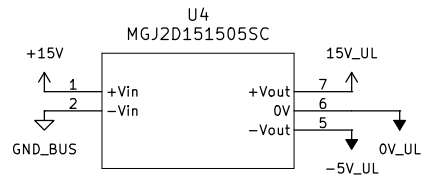
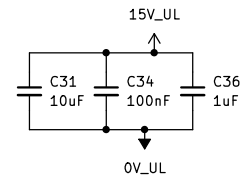
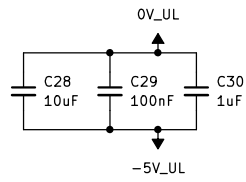
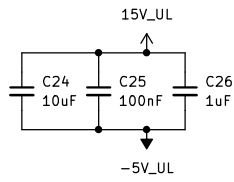
Formül uygulandığında:

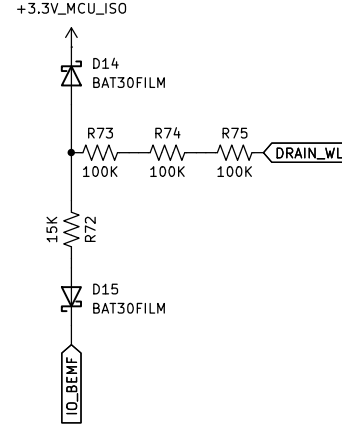
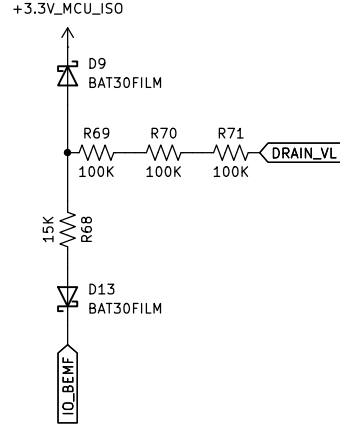
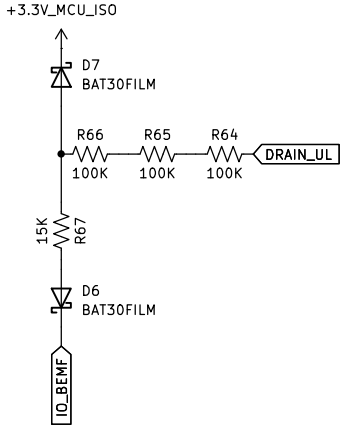
$$\begin{aligned} t_{bk} &= -C_{BLK}(R_{chg}) \cdot \ln\left(1 - \frac{V_{DESAT}}{V_{DD} + I_{CHG} \cdot R_{chg}}\right) \\ &= -147.5 \, ns \times \ln\left(1 - \frac{9.15}{14 + 0.5 \, mA \times 1475}\right) = 143 \, ns \end{aligned}$$

Sonuç

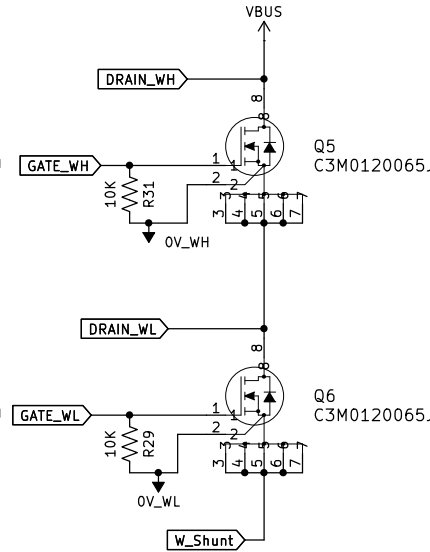
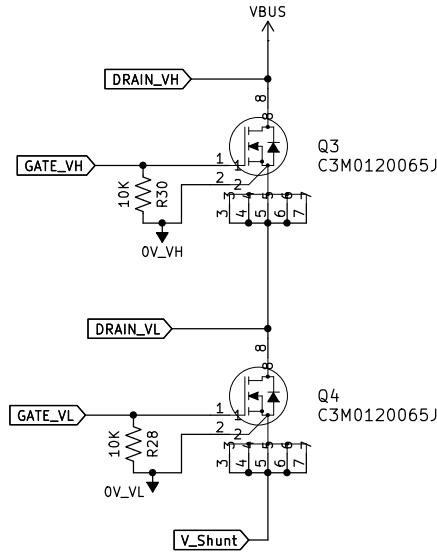
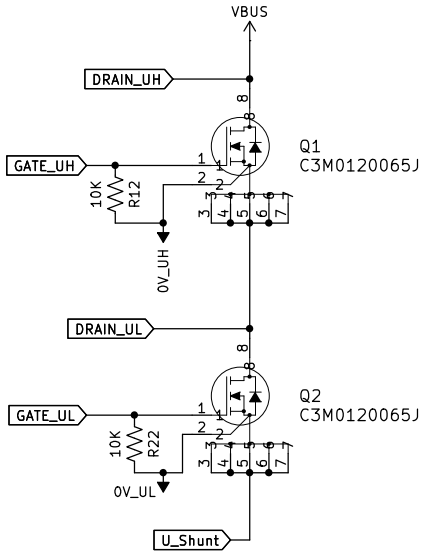
Aşama	Süre
RC eşiği aşma (ICHG'li formül)	~143 ns
RC eşiği aşma (LEB içi, ICHG=0)	~156 ns
Dahili LEB (algılamayı bloke eder)	200 ns (sabit)







!!!! BEMF tespit devresi izolasyonu bozar, motor parametreleri bilinmiyorsa kullanmak içindir !!!!
 Öneri: Parametre bulmak için Motoru 48V gibi düşük gerilimde çalıştır,
 Rs, Ls, pole pair, Ke → voltajdan bağımsız, aynı değerler çıkar
 J ve friction → aynı coast-down (serbest yavaşlama) yöntemi, düşük hızda da geçerli
 Bir kablo ile geçici olarak GND_BUS ve GND_MCU_ISO birleştir.



$310V \cdot I = (310 - 3.3) / 315k = 0.97mA$ □
 Her $100k\Omega$ görür: $98V < 200V$ (1206) □
 Güç: $(0.97mA)^2 \times 100k = 0.094W < 0.25W$ □
 $R_{bottom} / (R_{top} + R_{bottom}) = 15 / (300 + 15) = 15 / 315 = 1/21 \approx 0.04762$
 $R_{top} / R_{bottom} = 300 / 15 = 20$
 $R_{top}/R_{bottom} = (V_{BEMF_peak} / V_{ADC_hedef}) - 1$
 $= (69V / 3.3V) - 1$
 $= 20.91 - 1$
 $= 19.91 \approx 20$ □
 $V_{ADC} = 69 \times (1/21) = 69 / 21 = 3.286V \approx 3.3V$ □

BEMF klamp noktası: $3.3 \times 315 / 15 = 69V$
 → 69V altı BEMF doğrudan okunur
 → 69V üstü diyot klamlar, zarar vermez

Motorun Ke'si bilinmediği için hangi hızda 69V'u geçeceğini de bilmiyoruz
 1. Durum: ADC sinyali çok zayıf ise, R_{bottom} yükselt
 2. Durum: Motor BEMF beklenenden büyük ise R_{Bottom} küçült,
 saturasyona gireliyor olabilir de

Eğer Ke biliyor olsaydık
 $V_{BEMF} = Ke [V/rpm] \times N_{tanımlama} [rpm]$
 $Ke = 0.05 V/rpm$ (örnek)
 $N_{tespit} = 500 rpm$
 $V_{BEMF_peak} = 0.05 \times 500 = 25V$
 $ADC \text{ max } 3.3V, \%80 \text{ kullanırsak hedef } 2.5V:$
 $Oran = V_{hedef} / V_{BEMF} = 2.5V / 25V = 0.1$

$R_{bottom} / (R_{top} + R_{bottom}) = 0.1$
 $R_{top} / R_{bottom} = 9$

→ $R_{bottom} = 10k\Omega$, $R_{top} = 90k\Omega$ ($2 \times 47k\Omega$ seri)

Kısacası formül
 $R_{top}/R_{bottom} = (V_{BEMF_peak} / V_{ADC_hedef}) - 1$
 69V klamp noktasını makul bir marjın olarak seçtik.

Osiloskop ile Ke bulma:

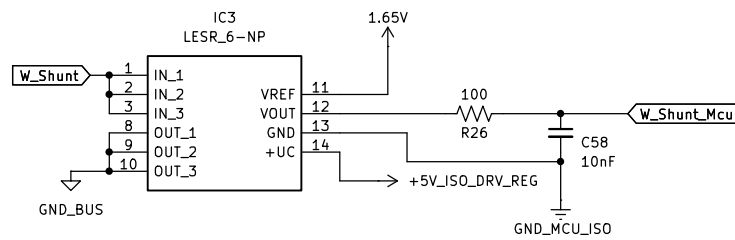
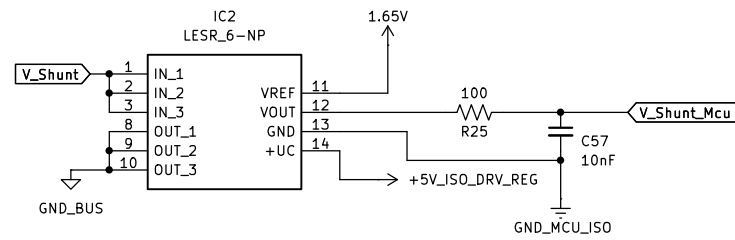
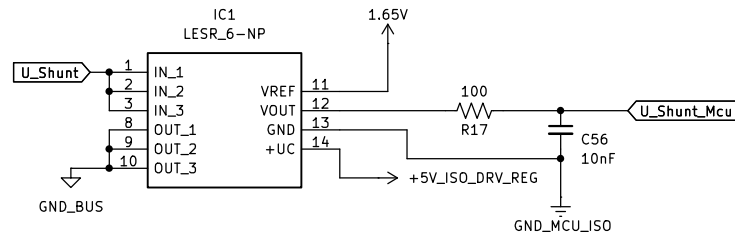
Motor serbest dönsün (sürücüyü bağlama)
 Osiloskop → Herhangi bir fazı ölç → Sinüs dalga görürsün
 $Ke [V_{peak}/rpm] = V_{peak} / N_{rpm}$
 veya
 $Ke [V/(rad/s)] = V_{peak} / \omega$ ($\omega = 2\pi \times N_{rpm} / 60$)
 Hız bilgisi için: Frekansı osiloskoptan oku → $N = f_{elektrik} \times 60 / (\text{pole_pairs})$
 yada doğrudan belirli bir rpm ile döndür.

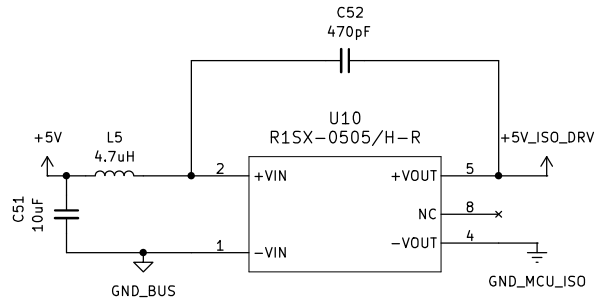
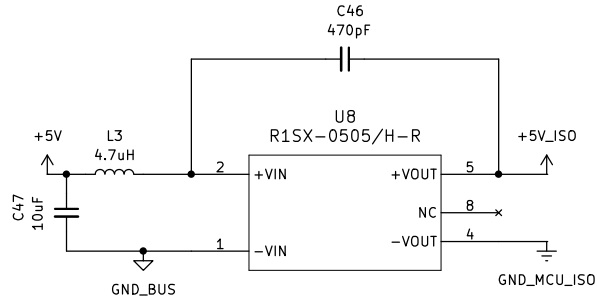
$$17 + (10000/2000) \times 3 = 66\text{mA} \rightarrow 66\text{mA} \times 5\text{V} = 330\text{mW}$$

$$100\Omega + 10\text{nF} \rightarrow f_c = 159\text{kHz}$$

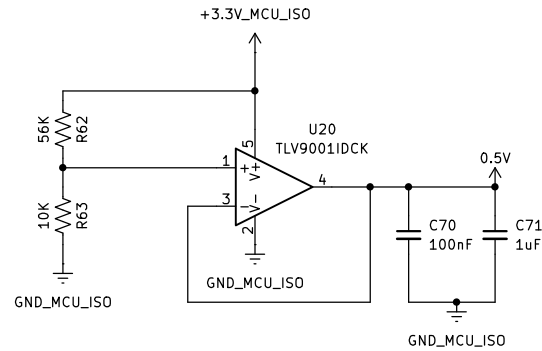
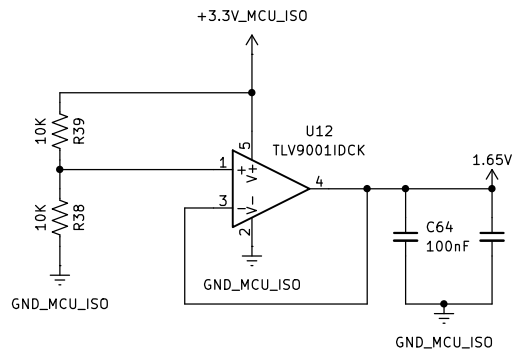
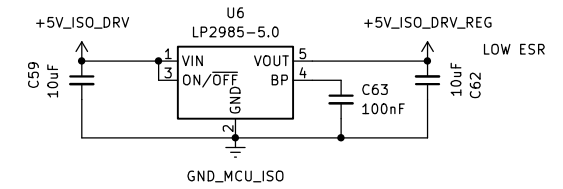
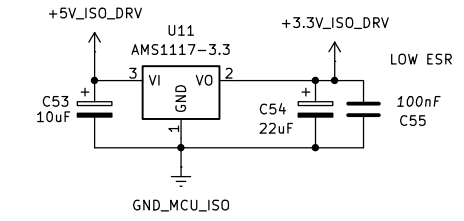
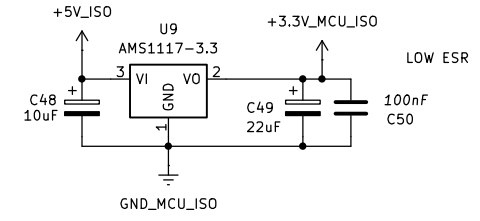
Gain: 104,2

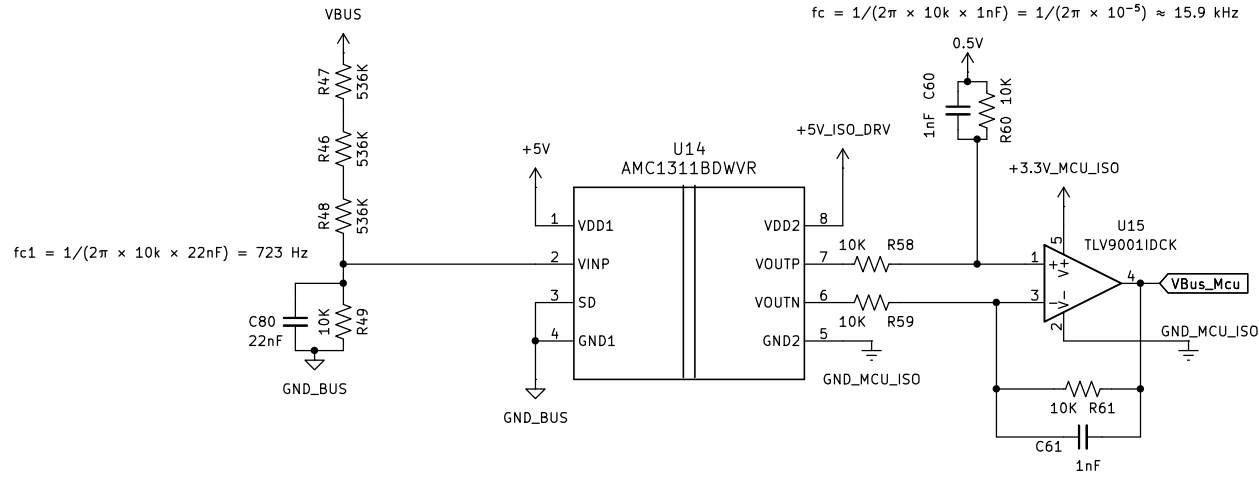
$$\begin{aligned} &= 1 / (2\pi \times 3300 \times 330 \times 10^{-12}) \\ &= 1 / (2\pi \times 1.089 \times 10^{-6}) \\ &= 1 / 6.84 \times 10^{-6} \\ &f_c \approx 146 \text{ kHz} \end{aligned}$$





GÜÇ Hesabı tekrar yapılacak





VBus ölçümü için FOC parametre değeri
 $(3 \times 536k + 10k) / 10k = 1618k / 10k = 161.8 \text{ V/V}$

ADC1,2,3 -> shunt
ADC4 -> VBus

